Eco-Share – Campus Carpooling & Bike Sharing System

1. Introduction

Daily commuting to college is a common challenge faced by students. Many rely on overcrowded public transport or expensive private vehicles, which leads to inconvenience, increased costs, and environmental harm. In many cases, students also face issues such as delays, lack of availability, and safety concerns during travel.

With the rise of digital platforms, there is an opportunity to simplify commuting through shared mobility solutions within a trusted campus environment. Eco-Share is designed as a smart solution to enable students to coordinate rides efficiently and safely while promoting sustainability.

Key concerns addressed:

- Overcrowded and unreliable public transport
- High fuel and travel costs
- Increased traffic and pollution
- Lack of a trusted ride-sharing system within campuses

2. Description

Eco-Share is a campus-based carpooling and bike-sharing platform that allows students to share rides with peers traveling in the same direction. It provides a structured and secure way for students to coordinate travel without relying on informal communication channels.

The platform enables users to:

- Post ride details (origin, destination, time)
- View available rides in real time
- Join rides based on seat availability
- Track active and upcoming trips

The system ensures a centralized and organized way of managing daily travel, replacing informal communication methods such as messaging groups or notice boards. It also enhances convenience by providing all travel-related information in one place.

3. Problem Statement

Public transportation to colleges can be crowded or unreliable, and individual commuting via private vehicles is expensive and environmentally unfriendly. Students often face difficulties in finding reliable and cost-effective travel options on a daily basis.

Currently, there is no safe, campus-restricted platform that allows students to coordinate shared travel effectively and securely. Existing solutions are either:

- Not limited to campus users (lack of trust)
- Unorganized and scattered across different platforms
- Inefficient in managing seat availability and ride coordination

This creates a need for a dedicated system that ensures safety, reliability, and efficiency in student commuting.

4. Scenario

A student traveling from home to college can post a ride by entering details such as pickup location, destination, and time. The student also specifies the number of available seats for other passengers.

Other students with similar routes can:

- View the ride on the platform
- Check seat availability
- Request or directly join the ride

The system dynamically updates the number of available seats, ensuring proper coordination and preventing overbooking. All rides are displayed on a “Live Trips” board, making it easy for users to find suitable options.

Example Flow:

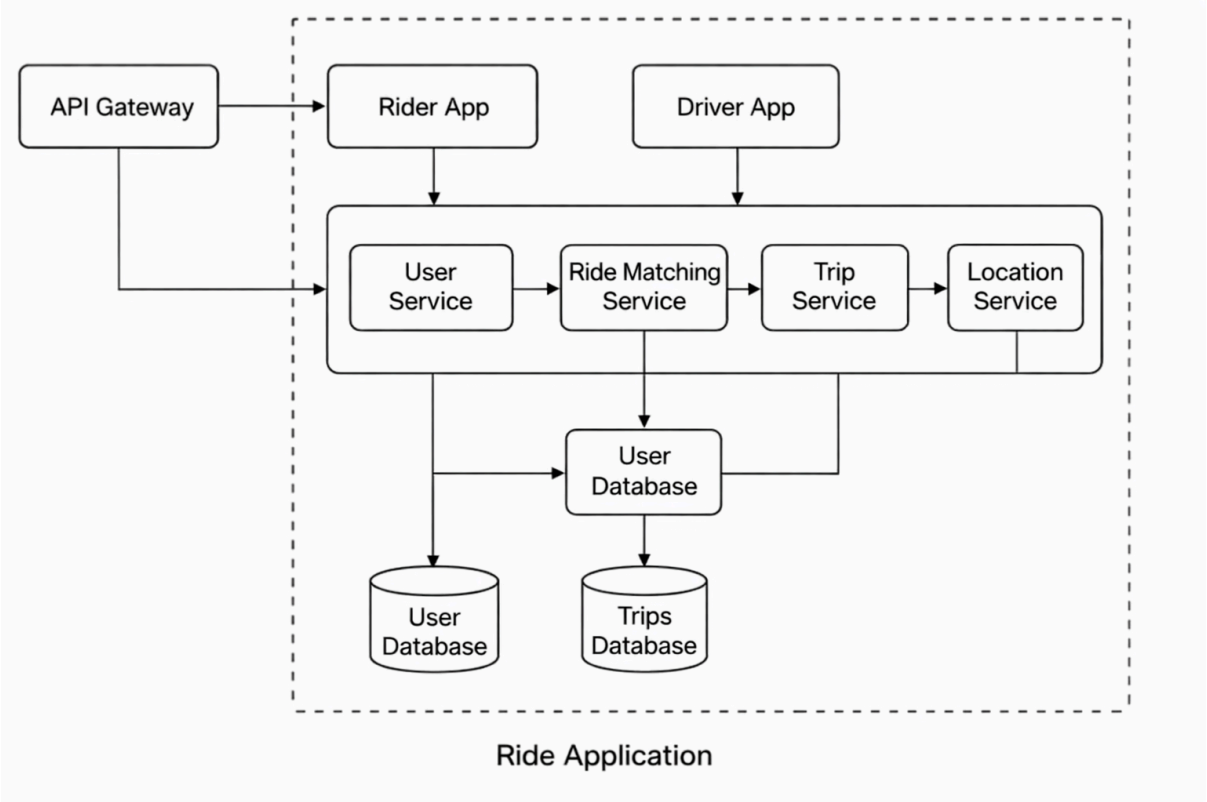
- Student A posts a ride with 3 available seats
- Student B and C join the ride
- Seat count updates from 3 → 1
- Ride becomes unavailable once seats are filled

This approach:

- Reduces travel costs by sharing expenses
- Minimizes the number of vehicles on the road
- Promotes eco-friendly commuting
- Builds a sense of community within the campus

5. Architecture

The Eco-Share system follows a three-tier architecture to ensure scalability, security, and efficient performance.



1. Presentation Layer (Frontend)

This layer provides the user interface developed using React.js. It allows users to register, log in, post rides, view live trips, and join rides. The interface dynamically displays real-time updates such as seat availability and active trips.

2. Application Layer (Backend)

The backend manages all business logic, including ride creation, joining functionality, and seat updates. It also handles authentication and role-based access control, ensuring that only users with valid institutional email addresses can access the platform. This layer processes requests and communicates with the database.

3. Data Layer (Database)

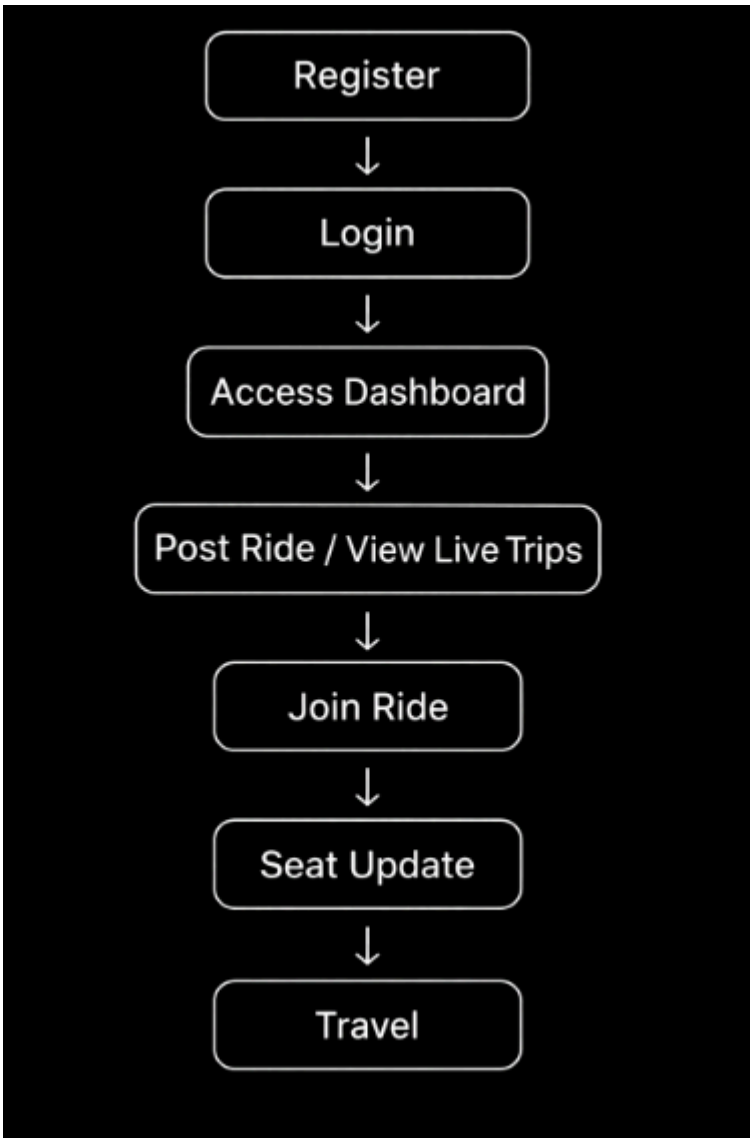
The database stores all system-related data, including user details, ride information, and booking records. It ensures data consistency, integrity, and efficient retrieval of information.

6. Project Flow (System Flow)

1. User registers using a college email address
2. User logs into the system
3. User can post a ride or view available rides
4. Other users join rides based on availability
5. Seat count updates automatically
6. Ride details are displayed on the Live Trips board

7. User Flow

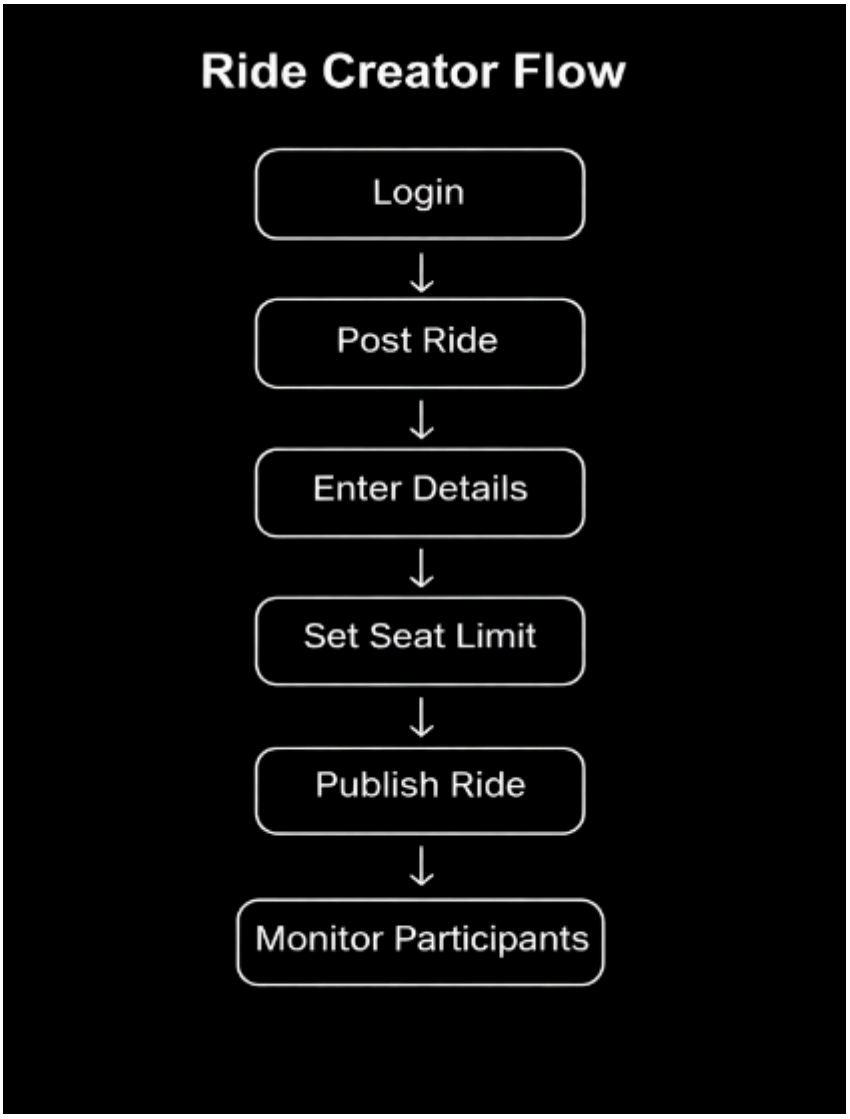
Student/User Flow



Explanation:

- The user registers using a valid college email address.
- After login, the user accesses the dashboard.
- The user can either post a ride or view available trips.
- The user joins a ride based on availability.
- The system updates the seat count automatically.

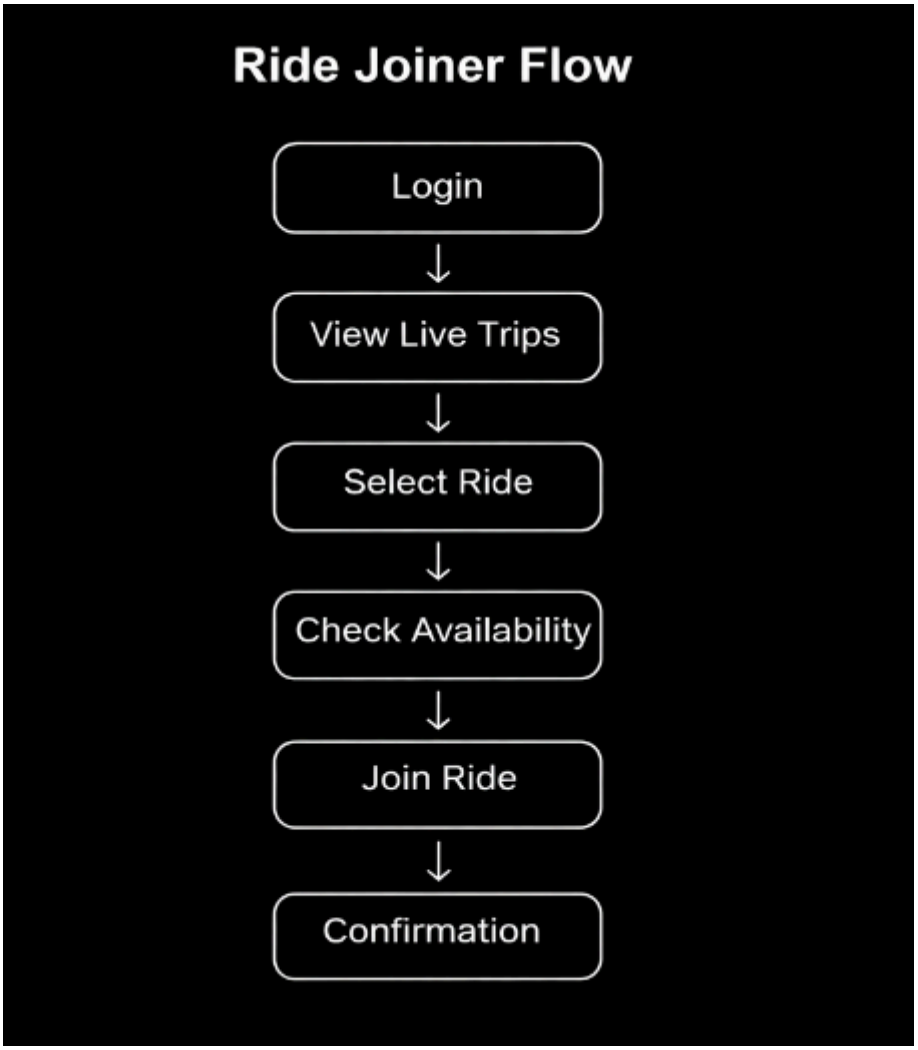
Ride Creator Flow



Explanation:

- The user creates a ride by entering travel details.
- The number of seats is defined.
- The ride is published for others to join.
- The creator monitors participants.

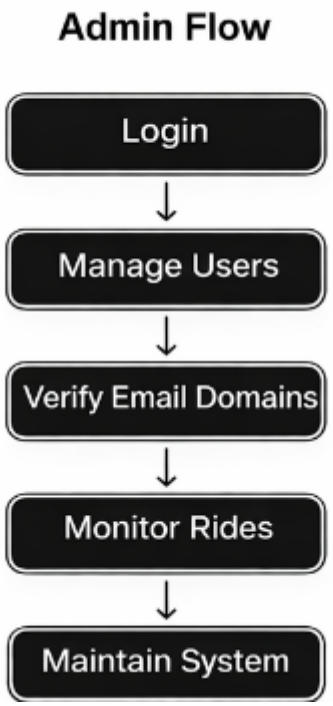
Ride Joiner Flow



Explanation:

- The user browses available rides.
- The user selects a ride and checks seat availability.
- If seats are available, the user joins the ride and receives confirmation.

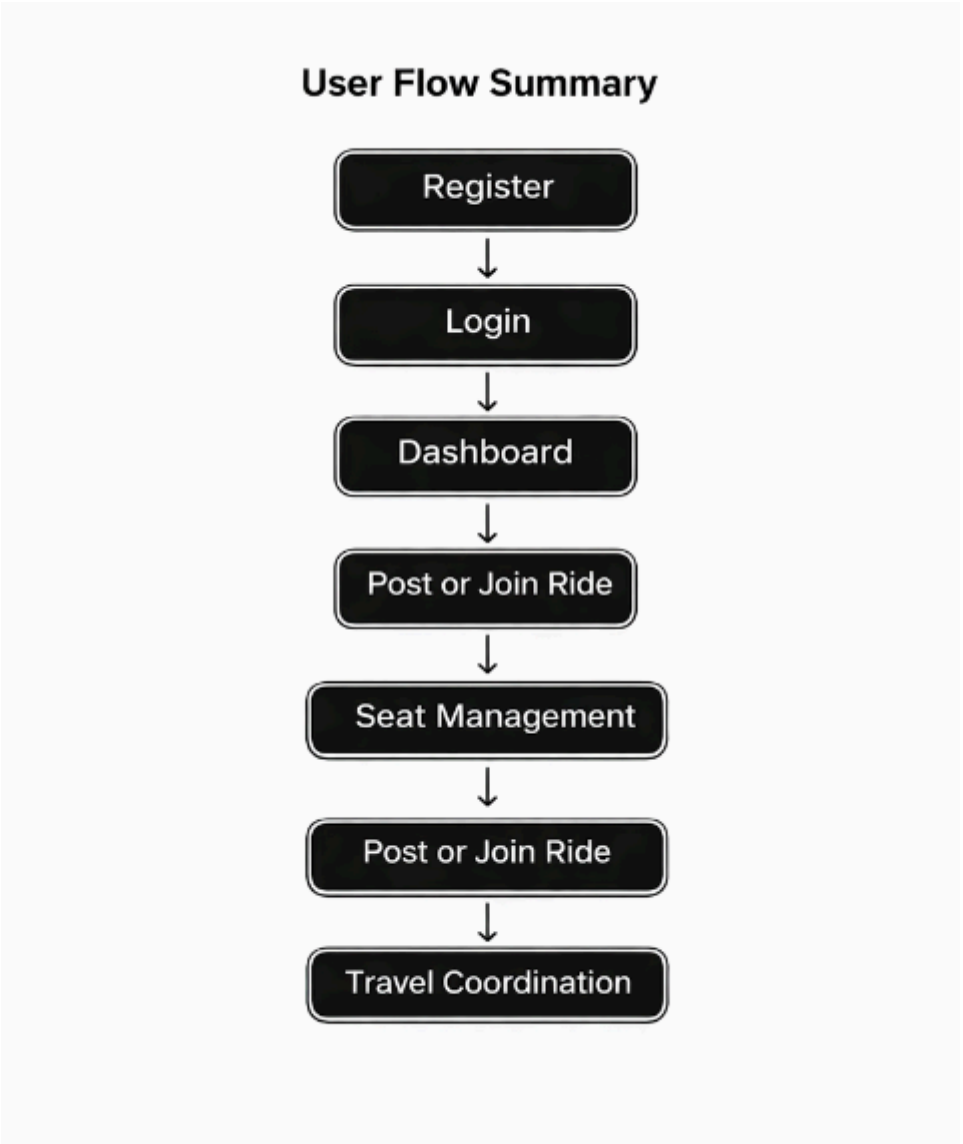
Admin Flow



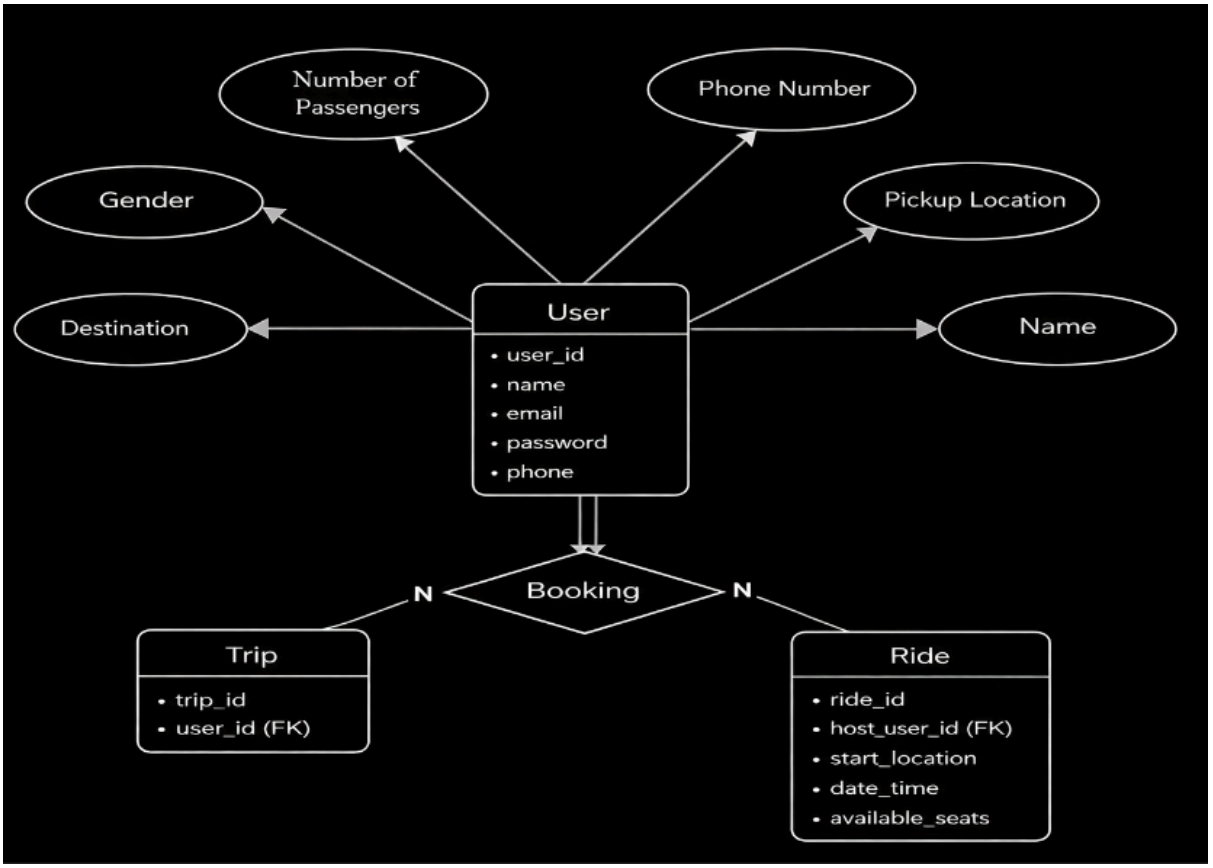
Explanation:

- The admin manages users and ensures valid registrations.
- The admin monitors rides and system activities.
- Necessary actions are taken to maintain system integrity.

User Flow Summary



ER Diagram



8. Pre-Requisites

- Basic knowledge of HTML, CSS, and JavaScript
- Understanding of React.js
- Knowledge of backend development and APIs
- Familiarity with database concepts

9. Required Technologies

- Frontend: React.js
- Backend: Node.js / Express

- Database: MongoDB
- Authentication: Email-based verification
- Deployment: Replit / Render / Vercel

10. Suggested Database

- **Users Collection:** Name, Email, Role
- **Rides Collection:** Origin, Destination, Time, Available Seats
- **Bookings Collection:** Ride ID, User ID, Seat Status

11. Key Features

- Post a Ride with origin, destination, and time
- Real-time Seat Counter
- Live Trips Board for daily travel options
- Secure login using institutional email
- Easy and efficient ride joining system

12. Optional Advanced Features

- Email notifications for ride updates
- GPS-based route optimization

- Ride history tracking
- User rating and feedback system
- Integration of bike-sharing module

13. Learning Outcomes

- Understanding real-world problem solving
- Hands-on experience in full-stack development
- Knowledge of authentication and access control
- Database design and data management
- Development of scalable and user-friendly applications